Statistics, Data Analysis & Sampling

ClipSAT Printable Practice Suite (clipsat.org) • Topic 9.01 • 20 FRQs + 25 MCQs • Difficulty Progression • Cross-Lesson Synthesis

ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • [FRQ] • Target: Measures of Center & Spread

Compute the mean, median, range, and interquartile range (IQR) of the dataset $\{12, 15, 15, 18, 22, 25, 33\}$.

Sum $= 12 + 15 + 15 + 18 + 22 + 25 + 33 = 140$, so Mean $= \frac{140}{7} = 20$. The middle (4th) value is Median $= 18$. Range $= 33 - 12 = 21$. Lower half $\{12, 15, 15\}$ has $Q_1 = 15$; upper half $\{22, 25, 33\}$ has $Q_3 = 25$; thus IQR $= 25 - 15 = 10$.

2 [Foundational] • [FRQ] • Target: Target Mean & Weighted Totals

A student has a mean score of 84 across 5 exams. What score must the student earn on the 6th exam to raise the overall mean to 86?

Total points on the first 5 exams $= 5 \times 84 = 420$. Required total on 6 exams $= 6 \times 86 = 516$. Thus the 6th exam score must be $516 - 420 = 96$.

3 [Foundational] • [FRQ] • Target: Comparing Boxplots (Median & IQR)

The side-by-side boxplots below compare daily commuter times (in minutes) on Route A and Route B. State the median and IQR for each route, and identify which route has more consistent travel times.

Route A has five-number summary $(15, 22, 28, 34, 45)$, so Median $= 28\,\text{min}$ and IQR $= 34 - 22 = 12\,\text{min}$. Route B has $(10, 20, 32, 42, 50)$, so Median $= 32\,\text{min}$ and IQR $= 42 - 20 = 22\,\text{min}$. Route A has smaller IQR and range, so its travel times are more consistent.

4 [Foundational] • [FRQ] • Target: Two-Way Tables & Conditional Probability

In a clinical study of 200 patients, 120 received Treatment X (90 improved, 30 did not) and 80 received a Placebo (30 improved, 50 did not). Compute (a) $P(\text{Improved})$, (b) $P(\text{Improved} \mid \text{Treatment X})$, and (c) $P(\text{Treatment X} \mid \text{Improved})$.

Total improved $= 90 + 30 = 120$. (a) $P(\text{Improved}) = \frac{120}{200} = 0.60$. (b) $P(\text{Improved} \mid \text{Treatment X}) = \frac{90}{120} = 0.75$. (c) $P(\text{Treatment X} \mid \text{Improved}) = \frac{90}{120} = 0.75$.

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5 [Foundational] • [FRQ] • Target: Scatterplots, Line of Best Fit & Residuals

A scatterplot of study hours x versus quiz score y has line of best fit $\hat{y} = 4.5x + 52$, as shown. Calculate the predicted score for $x = 6$ hours and the residual for a student who studied 6 hours and scored 83.

Predicted score: $\hat{y}(6) = 4.5(6) + 52 = 27 + 52 = 79$. Residual $= y - \hat{y} = 83 - 79 = +4$ (the student scored 4 points above the model prediction).

6 [Foundational] • [FRQ] • Target: Conceptual Understanding of Standard Deviation

Explain without calculation which dataset has the larger standard deviation: Dataset A $= \{10, 12, 14, 16, 18\}$ or Dataset B $= \{10, 10, 14, 18, 18\}$. Then verify by computing both sample variances.

Both datasets have mean $\bar{x} = 14$, but Dataset B places four of its five values at the extreme endpoints 10 and 18 (distance 4 from the mean), whereas Dataset A has two values at distance 2 and two at distance 4. Indeed, $\sum(x_i - 14)^2 = 40$ for A vs. 64 for B, so Dataset B has a larger standard deviation.

7 [Foundational] • [FRQ] • Target: Margin of Error & Confidence Intervals

A public-opinion poll of 600 randomly selected registered voters in a state of $4{,}000{,}000$ registered voters finds that 54% support Proposition 7, with a margin of error of $\pm 4\%$ at the 95% confidence level. State the plausible interval for the true state-wide support percentage and the corresponding interval of voter counts.

The plausible percentage interval is $54\% \pm 4\% = [50\%, 58\%]$. Multiplying by $4{,}000{,}000$ gives an interval of $[2{,}000{,}000,\ 2{,}320{,}000]$ voters.

Part I — Free-Response Questions (FRQs): Intermediate Modeling (8–14)

8 [Intermediate] • [FRQ] • Target: Sensitivity of Mean vs. Median to Outliers

A dataset of 9 distinct positive integers has median 18 and mean 20. If the largest number in the dataset is increased by 18 while all other numbers remain unchanged, what are the new mean and new median?

Increasing the maximum value by 18 increases the sum of the 9 numbers by 18, so the new mean is $20 + \frac{18}{9} = 22$. Because the rank order of the 9 values does not change, the 5th value (the median) remains 18.

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9 [Intermediate] • [FRQ] • Target: Mean & Median from Frequency Distributions

A frequency table shows the number of books read over the summer by 25 students: 1 book (4 students), 2 books (7 students), 3 books (6 students), 4 books (5 students), 5 books (3 students). Find the mean and median number of books read.

Total books $= 1(4) + 2(7) + 3(6) + 4(5) + 5(3) = 4 + 14 + 18 + 20 + 15 = 71$, so Mean $= \frac{71}{25} = 2.84$. The median is the 13th value in sorted order; cumulative frequencies are $4, 11, 17$, so the 13th student falls in the '3 books' group (Median $= 3$).

10 [Intermediate] • [FRQ] • Target: Sampling Without Replacement

A box contains 5 red marbles and 3 blue marbles. Two marbles are drawn at random without replacement. Find the probability that (a) both marbles are red, and (b) the two marbles have different colors.

(a) $P(\text{RR}) = \frac{5}{8} \cdot \frac{4}{7} = \frac{20}{56} = \frac{5}{14}$. (b) $P(\text{different}) = P(\text{RB}) + P(\text{BR}) = \frac{5}{8} \cdot \frac{3}{7} + \frac{3}{8} \cdot \frac{5}{7} = \frac{30}{56} = \frac{15}{28}$.

11 [Intermediate] • [FRQ] • Target: Sample Size & Margin of Error Scaling

Explain how the margin of error of a sample proportion $\text{ME} \approx 1.96\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$ changes if the sample size n is quadrupled from $n = 250$ to $n = 1{,}000$.

Because margin of error is inversely proportional to $\sqrt{n}$, multiplying the sample size n by 4 divides the margin of error by $\sqrt{4} = 2$ (cutting the margin of error in half).

12 [Intermediate] • [FRQ] • Target: Observational Studies vs. Randomized Experiments

Determine whether each study design supports a cause-and-effect conclusion or only an association, and identify the target population to which results can be generalized: (a) A random sample of 500 adults in City X is surveyed on sleep duration and blood pressure. (b) 200 volunteers with hypertension are randomly assigned to either a daily walking regimen or a control group.

(a) Observational study with random sampling: establishes correlation/association (not causation), generalizable to all adults in City X. (b) Randomized controlled experiment on volunteers: establishes a causal effect of the walking regimen, strictly applicable to individuals similar to the study volunteers.

13 [Intermediate] • [FRQ] • Target: Outlier Detection via the 1.5-IQR Rule

Using the 1.5-IQR rule, determine whether the minimum value (2) or maximum value (64) is an outlier in a dataset with $Q_1 = 20$, median $= 27$, and $Q_3 = 36$.

Here $\text{IQR} = 36 - 20 = 16$, so $1.5 \times \text{IQR} = 24$. The lower fence is $Q_1 - 24 = 20 - 24 = -4$ (since $2 \geq -4$, 2 is not an outlier). The upper fence is $Q_3 + 24 = 36 + 24 = 60$ (since $64 > 60$, 64 is a high outlier).

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14 [Intermediate] • [FRQ] • Target: Weighted Means Across Subgroups

Class A (20 students) has an exam mean of 78, and Class B (30 students) has an exam mean of 88. Find the combined mean of all 50 students.

Combined mean is the weighted average $\bar{x} = \frac{20(78) + 30(88)}{20 + 30} = \frac{1560 + 2640}{50} = \frac{4200}{50} = 84$.

Part I — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • [FRQ] • Target: Linear Transformations of Random Variables & Data • Cross-Lesson: Topic 1.01 (Linear Transformations)

Every observation x_i in a dataset with mean $\bar{x} = 25$ and standard deviation $s_x = 4$ is transformed according to the linear rule $y_i = 3x_i - 15$. Find the mean $\bar{y}$, standard deviation s_y, and variance s_y^2 of the transformed dataset.

By linearity of expectation, $\bar{y} = 3\bar{x} - 15 = 3(25) - 15 = 60$. Standard deviation scales by $|3|$ and ignores shifts: $s_y = 3(4) = 12$. Variance scales by 3^2: $s_y^2 = 12^2 = 144$.

16 [Challenging] • [FRQ] • Target: Residual Plot Diagnostics for Nonlinearity • Cross-Lesson: Topic 3.01 (Quadratic Models)

A residual plot $(x_i, y_i - \hat{y}_i)$ from a linear regression displays a pronounced U-shaped parabolic pattern (positive residuals at both ends, negative residuals in the middle). Explain what this indicates about the appropriateness of the linear model and what model family would be more appropriate.

A deterministic U-shaped pattern in the residuals indicates that the underlying relationship between x and y is nonlinear (specifically concave upward), so a linear model systematically underpredicts at the extremes and overpredicts in the middle. A quadratic (or exponential) regression model is more appropriate.

17 [Challenging] • [FRQ] • Target: Extremal Optimization with Mean, Median & Mode Constraints • Cross-Lesson: Topic 2.01 (Inequalities & Optimization)

Five positive integers have a mode of 4, a median of 6, and a mean of 7. What is the largest possible value of the maximum integer in the list?

Order the integers $a \le b \le c \le d \le e$ with $a + b + c + d + e = 35$ and $c = 6$. Since the mode is 4 (which is less than 6), we must have $a = b = 4$. To maximize e, make d as small as possible without creating another mode ($d = 7$, since $d = 6$ would make 6 tied for mode). Then $4 + 4 + 6 + 7 + e = 35 \Rightarrow e = 14$.

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18 [Challenging] • [FRQ] • Target: Bayes' Theorem & Law of Total Probability • Cross-Lesson: Topic 5.01 (Weighted Proportions)

In a factory, Machine 1 produces 60% of all microchips (with a 2% defect rate) and Machine 2 produces 40% of all microchips (with a 5% defect rate). If a randomly selected microchip is found to be defective, what is the probability that it was produced by Machine 2?

By Bayes' Theorem: $P(D) = 0.60(0.02) + 0.40(0.05) = 0.012 + 0.020 = 0.032$. Therefore $P(M_2 \mid D) = \frac{0.020}{0.032} = \frac{20}{32} = \frac{5}{8} = 0.625$.

19 [Challenging] • [FRQ] • Target: Piecewise Median-Mean Equations • Cross-Lesson: Topic 1.01 (Piecewise Linear Equations)

A dataset of 6 values is $\{4, 7, 9, 12, 15, x\}$. For what value(s) of x is the mean of the dataset equal to its median?

Sum $= 47 + x$, so Mean $= \frac{47+x}{6}$. Case 1 ($x \le 9$): Median is $\frac{7+9}{2} = 8 \Rightarrow 47 + x = 48 \Rightarrow x = 1 \le 9$ (valid). Case 2 ($9 < x < 12$): Median is $\frac{9+x}{2} = \frac{47+x}{6} \Rightarrow 27 + 3x = 47 + x \Rightarrow x = 10 \in (9, 12)$ (valid). Case 3 ($x \ge 12$): Median is $\frac{9+12}{2} = 10.5 \Rightarrow 47 + x = 63 \Rightarrow x = 16 \ge 12$ (valid). Thus $x \in \{1, 10, 16\}$.

20 [Challenging] • [FRQ] • Target: Geometric Probability on Concentric Circles • Cross-Lesson: Topic 9.01 (Area of Circles & Annuli)

A dartboard consists of an outer circle of radius $10\,\text{cm}$ and a concentric inner bullseye circle of radius $4\,\text{cm}$. If a dart lands uniformly at random on the board, find the probability that it lands in the outer ring (outside the bullseye).

Area of outer circle $= 100\pi\,\text{cm}^2$; area of inner bullseye $= 16\pi\,\text{cm}^2$. Area of outer ring $= 100\pi - 16\pi = 84\pi\,\text{cm}^2$. Probability $= \frac{84\pi}{100\pi} = \frac{21}{25} = 0.84$.

Part II — Multiple-Choice Questions (MCQs): Foundational Concepts (21–28)

21 [Foundational] • [MCQ] • Target: Calculating the Median

What is the median of the data set $\{18, 7, 24, 11, 15, 30, 12\}$?

a. 12

b. 15 ✓

c. 16.7

d. 18

Sort the 7 numbers in ascending order: $7, 11, 12, 15, 18, 24, 30$. The middle (4th) number is 15.

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22 [Foundational] • [MCQ] • Target: Basic Event Probability

A fair six-sided die is rolled once. What is the probability of rolling a number that is both prime and odd?

a. $\frac{1}{6}$

b. $\frac{1}{3}$ ✓

c. $\frac{1}{2}$

d. $\frac{2}{3}$

The prime outcomes on $\{1, 2, 3, 4, 5, 6\}$ are $\{2, 3, 5\}$, of which $\{3, 5\}$ are odd (2 outcomes). Thus $P = \frac{2}{6} = \frac{1}{3}$.

23 [Foundational] • [MCQ] • Target: Finding a Missing Value from the Mean

If the mean of $4, 9, 11, 15,$ and x is 12, what is the value of x?

a. 17

b. 19

c. 21 ✓

d. 23

Since the mean of 5 numbers is 12, their sum is $5 \times 12 = 60$. Thus $4 + 9 + 11 + 15 + x = 39 + x = 60 \Rightarrow x = 21$.

24 [Foundational] • [MCQ] • Target: Resistant vs. Non-Resistant Statistics

Which statistic is LEAST affected when a single extreme high outlier is added to a dataset of 50 home prices?

a. Mean

b. Range

c. Standard deviation

d. **Median** ✓

The median depends only on the middle rank(s) of the sorted data and is resistant to extreme outliers, whereas the mean, range, and standard deviation all increase substantially.

25 [Foundational] • [MCQ] • Target: Addition Rule of Probability

In a survey of 150 high school seniors, 90 plan to study science, 60 plan to study engineering, and 30 plan to study both. If one senior is selected at random, what is the probability that the senior plans to study science or engineering?

a. 0.60

b. 0.75

c. 0.80 ✓

d. 1.00

By the addition rule, $n(S \cup E) = 90 + 60 - 30 = 120$. Probability $= \frac{120}{150} = 0.80$.

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26 [Foundational] • [MCQ] • Target: Interpreting the Correlation Coefficient r

A scatterplot shows a strong negative linear association between x and y. Which of the following could be the sample correlation coefficient r?

a. $r = -0.92$ ✓

b. $r = -0.15$

c. $r = 0.08$

d. $r = 0.89$

A strong negative linear association corresponds to a correlation coefficient r close to -1, such as $r = -0.92$.

27 [Foundational] • [MCQ] • Target: Reading Mode from a Frequency Histogram/Bar Chart

The bar chart below shows the distribution of scores on a 5-point quiz. What is the mode of the quiz scores?

a. 2

b. 3

c. 4 ✓

d. 5

Score 4 has the highest frequency (14 students), so the mode is 4.

28 [Foundational] • [MCQ] • Target: Random Sampling & Bias Prevention

A researcher wants to estimate the average daily screen time of all $2{,}400$ students at a high school. Which sampling method is most likely to yield an unbiased estimate?

a. Surveying 100 students in the computer club.

b. **Selecting 100 students using a random number generator from the full school roster.** ✓

c. Posting an optional online poll on the school website.

d. Asking the first 100 students who arrive at the library before 8:00 AM.

A simple random sample from the complete population frame gives every student an equal probability of selection and avoids voluntary-response or convenience bias.

Part II — Multiple-Choice Questions (MCQs): Intermediate Exam Practice (29–37)

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29 [Intermediate] • [MCQ] • Target: Shift Invariance of Standard Deviation

Set X is $\{2, 4, 6, 8, 10\}$ and Set Y is $\{102, 104, 106, 108, 110\}$. Which statement correctly compares their standard deviations?

a. Set X has a greater standard deviation than Set Y.

b. Set Y has a greater standard deviation than Set X.

c. **Both sets have the exact same standard deviation.** ✓

d. The standard deviation of Set Y is 100 plus the standard deviation of Set X.

Adding a constant ($+100$) to every element shifts the mean by $+100$ but leaves all deviations from the mean unchanged, so both sets have the exact same standard deviation.

30 [Intermediate] • [MCQ] • Target: Statistical Independence in Two-Way Tables

In a two-way table of 280 voters classified by age group (Under 40 vs. 40+) and opinion (Support vs. Oppose), 120 voters are Under 40, and 75 of those Under 40 support the measure. If opinion is statistically independent of age group, how many of the 160 voters aged 40+ support the measure?

a. 90

b. 96

c. 100 ✓

d. 112

Among voters Under 40, the support proportion is $\frac{75}{120} = \frac{5}{8} = 62.5\%$. By statistical independence, the same proportion applies to the 160 voters aged 40+: $\frac{5}{8} \times 160 = 100$.

31 [Intermediate] • [MCQ] • Target: Residuals in Nonlinear Regression

A exponential regression model $\hat{y} = 25(1.4)^x$ is fit to a dataset. If the actual data point at $x = 2$ is $(2, 46.5)$, what is the residual at $x = 2$?

a. -2.5 ✓

b. -1.5

c. 1.5

d. 2.5

Predicted value at $x = 2$: $\hat{y} = 25(1.4)^2 = 25(1.96) = 49.0$. Residual $= y - \hat{y} = 46.5 - 49.0 = -2.5$.

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32 [Intermediate] • [MCQ] • Target: Effect of Linear Scaling on Median & IQR

A dataset of 21 integers has median 30 and interquartile range 14. If every number in the dataset is multiplied by 2 and then increased by 5, what are the new median and new IQR?

a. Median $= 65$, IQR $= 28$ ✓

b. Median $= 65$, IQR $= 33$

c. Median $= 60$, IQR $= 28$

d. Median $= 60$, IQR $= 33$

Measures of position (median) scale and shift: $2(30) + 5 = 65$. Measures of spread (IQR) scale by $|2|$ but are unaffected by the $+5$ shift: $2(14) = 28$.

33 [Intermediate] • [MCQ] • Target: Scope of Inference (Generalizability vs. Causation)

A clinical trial randomly assigns 400 adult migraine sufferers to receive either a new medication or a placebo, and finds a statistically significant reduction in migraine frequency in the medication group. To which group can this causal finding be safely generalized?

a. All adults worldwide

b. All children and adults with headaches

c. Adults suffering from migraines similar to those in the study ✓

d. Only the 400 participants in the study

Random assignment establishes causation, and results generalize to the population represented by the study's inclusion criteria (adult migraine sufferers similar to the participants).

34 [Intermediate] • [MCQ] • Target: Compound Probability Without Replacement

A bag contains 4 green, 5 yellow, and 6 white balls. If two balls are drawn successively without replacement, what is the probability that neither ball is white?

a. $\frac{6}{35}$

b. $\frac{12}{35}$ ✓

c. $\frac{9}{25}$

d. $\frac{19}{35}$

There are $4 + 5 = 9$ non-white balls out of 15 total. Thus $P(\text{neither white}) = \frac{9}{15} \cdot \frac{8}{14} = \frac{3}{5} \cdot \frac{4}{7} = \frac{12}{35}$.

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35 [Intermediate] • [MCQ] • Target: Empirical Rule & Symmetry

In a symmetric, bell-shaped distribution with mean 70 and standard deviation 8, approximately 68% of the data lies within one standard deviation of the mean, and 95% lies within two standard deviations. Approximately what percentage of the data lies between 70 and 86?

a. 34%

b. 47.5% ✓

c. 68%

d. 81.5%

The value 70 is the mean μ, and $86 = 70 + 2(8) = \mu + 2\sigma$. Since 95% of the distribution lies within $\mu \pm 2\sigma$, by symmetry half of that (47.5%) lies between μ and $\mu + 2\sigma$.

36 [Intermediate] • [MCQ] • Target: Subgroup Removal & Mean Reconstruction

The mean of 10 numbers is 15. When two numbers are removed, the mean of the remaining 8 numbers becomes 14. What is the average (arithmetic mean) of the two removed numbers?

a. 16

b. 17.5

c. 19 ✓

d. 21

Original sum $= 10 \times 15 = 150$. Remaining sum $= 8 \times 14 = 112$. Sum of the two removed numbers $= 150 - 112 = 38$, so their mean is $\frac{38}{2} = 19$.

37 [Intermediate] • [MCQ] • Target: Inverse-Square Sample Size Rule

A pollster wants to reduce the margin of error of a 95% confidence interval from ±6% to ±2%. By what factor must the random sample size n be multiplied?

a. 3

b. 6

c. 9 ✓

d. 12

To divide the margin of error by 3 (from 6% to 2%), the sample size n must be multiplied by $3^2 = 9$.

Part II — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • [MCQ] • Target: Minimizing Variance via Quadratic Optimization • Cross-Lesson: Topic 3.01 (Quadratic Minimization)

For a dataset of 4 numbers $\{2, 5, 9, x\}$, the sample variance is minimized when x equals the mean of the other three numbers. What is this value of x?

a. 5

b. $\frac{16}{3}$ ✓

c. 6

d. 7

The sum of squared deviations $\sum(x_i - \bar{x})^2$ as a quadratic function of x is minimized when x equals the mean of the fixed numbers: $x = \frac{2+5+9}{3} = \frac{16}{3}$.

39 [Challenging] • [MCQ] • Target: Independent Events & Quadratic Probability Equations • Cross-Lesson: Topic 3.01 (Quadratics)

Two independent events A and B satisfy $P(A) = p$, $P(B) = p + 0.2$, and $P(A \cap B) = 0.15$. What is the value of $P(A \cup B)$?

a. 0.45

b. 0.50

c. 0.65 ✓

d. 0.80

By independence,
$P(A \cap B) = p(p + 0.2) = 0.15 \Rightarrow p^2 + 0.2p - 0.15 = 0 \Rightarrow (p + 0.5)(p - 0.3) = 0 \Rightarrow p = 0.3$. Then $P(A) = 0.3, P(B) = 0.5$, and $P(A \cup B) = 0.3 + 0.5 - 0.15 = 0.65$.

40 [Challenging] • [MCQ] • Target: Binomial Random Walk on a Coordinate Line • Cross-Lesson: Topic 1.01 (Coordinate Lines)

In a coordinate grid, a random walker starts at $(0, 0)$ and takes 4 independent steps, each of length 1 unit either East $(+x)$ or North $(+y)$ with equal probability $\frac{1}{2}$. What is the probability that the walker ends on the line $y = x$?

a. $\frac{1}{4}$

b. $\frac{3}{8}$ ✓

c. $\frac{1}{2}$

d. $\frac{5}{16}$

After 4 steps, $x + y = 4$. To land on $y = x$, the walker must land at $(2, 2)$, which requires choosing exactly 2 East and 2 North steps out of 4: $\frac{\binom{4}{2}}{2^4} = \frac{6}{16} = \frac{3}{8}$.

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41 [Challenging] • [MCQ] • Target: OLS Slope Identity $m = r(s_y/s_x)$ & Centroid • Cross-Lesson: Topic 1.01 (Linear Equations)

The least-squares regression line for a bivariate dataset is $\hat{y} = -1.5x + 28$. If the mean of the y-values is $\bar{y} = 10$ and the ratio of standard deviations is $\frac{s_y}{s_x} = 2.0$, what is the correlation coefficient r and the mean $\bar{x}$?

a. $r = -0.75$ **and** $\bar{x} = 12$ ✓

b. $r = -0.75$ and $\bar{x} = 10$

c. $r = 0.75$ and $\bar{x} = 12$

d. $r = -0.50$ and $\bar{x} = 12$

Since slope $m = r\frac{s_y}{s_x}$, we have $-1.5 = r(2.0) \Rightarrow r = -0.75$. Using the centroid $(\bar{x}, \bar{y})$: $10 = -1.5\bar{x} + 28 \Rightarrow 1.5\bar{x} = 18 \Rightarrow \bar{x} = 12$.

42 [Challenging] • [MCQ] • Target: Minimax Statistical Constraint Optimization • Cross-Lesson: Topic 2.01 (Systems of Inequalities)

A list of 7 positive integers has median 10, unique mode 8, and mean 12. What is the minimum possible value of the largest integer in the list?

a. 15

b. 16

c. 17 ✓

d. 18

Sort $x_1 \le x_2 \le x_3 \le 10 \le x_5 \le x_6 \le x_7$ with sum $7 \times 12 = 84$. To minimize x_7, maximize x_1, x_2, x_3 while making 8 the unique mode: set $x_1 = x_2 = x_3 = 8$ (frequency 3). Then $x_5 + x_6 + x_7 = 84 - 34 = 50$. Splitting 50 as evenly as possible gives $16, 17, 17$ (where 17 has frequency $2 < 3$), so the minimum possible value of x_7 is 17.

43 [Challenging] • [MCQ] • Target: Geometric Probability in Inscribed Polygons • Cross-Lesson: Topic 8.01 (Polygons) & Topic 9.01 (Circles)

A point is chosen uniformly at random inside a regular hexagon of side length 2. What is the probability that the point lies inside the inscribed circle (which is tangent to all 6 sides)?

a. $\frac{\pi}{2\sqrt{3}}$ ✓

b. $\frac{\pi}{3\sqrt{3}}$

c. $\frac{\pi}{4\sqrt{3}}$

d. $\frac{2\pi}{3\sqrt{3}}$

For side $s = 2$, the hexagon area is $6 \cdot \frac{\sqrt{3}}{4}(2^2) = 6\sqrt{3}$, and the apothem (inradius) is $r = \sqrt{3}$, giving circle area $\pi(\sqrt{3})^2 = 3\pi$. The probability is $\frac{3\pi}{6\sqrt{3}} = \frac{\pi}{2\sqrt{3}}$.

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44 [Challenging] • [MCQ] • Target: Simpson's Paradox & Confounding Variables • Cross-Lesson: Topic 5.01 (Weighted Percentages)

Across two hospital clinics, Clinic A treats 90 mild cases (90% recovery) and 10 severe cases (50% recovery), while Clinic B treats 20 mild cases (95% recovery) and 80 severe cases (60% recovery). Which statement illustrates Simpson's Paradox?

a. Clinic A has a higher recovery rate in both categories, and a higher overall recovery rate.

b. **Clinic B has a higher recovery rate in both mild and severe categories, yet Clinic A has a higher overall combined recovery rate.** ✓

c. Both clinics have the exact same overall recovery rate.

d. Clinic B has a higher overall recovery rate despite lower category rates.

Clinic B outperforms Clinic A in both mild ($95\% > 90\%$) and severe ($60\% > 50\%$) cases, yet Clinic A's overall rate is $\frac{81+5}{100} = 86\%$ while Clinic B's overall rate is $\frac{19+48}{100} = 67\%$ because of confounding case-severity proportions (Simpson's Paradox).

45 [Challenging] • [MCQ] • Target: Linearized Regression Slope from Variances • Cross-Lesson: Topic 4.01 (Exponential Decay)

A exponential decay dataset (t, y) is transformed via $u = \ln(y)$. If the sample variance of t is $s_t^2 = 4$, the sample variance of u is $s_u^2 = 9$, and the points (t_i, u_i) lie exactly on a line with negative slope, what is the slope of that line?

a. -2.25

b. -1.5 ✓

c. -0.67

d. 1.5

Because the points lie exactly on a line with negative slope, $r = -1$. With $s_t = 2$ and $s_u = 3$, the slope is $m = r\frac{s_u}{s_t} = -1 \cdot \frac{3}{2} = -1.5$.

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